

Operating System (OS) – Medium Length Exam Answers for SPPU

These answers are written in simple language but with enough explanation to score marks in SPPU exams.

UNIT 1

1) Define Operating System and Explain its Functions

Definition

Operating System (OS) is system software that acts as an interface between the user and computer hardware. It controls and manages all hardware and software resources of the computer system.

Examples of Operating Systems are Windows, Linux, Android, and macOS.

Functions of Operating System

1. Process Management

Operating system manages execution of processes. It creates, schedules, and terminates processes.

2. Memory Management

OS allocates and deallocates memory to programs. It also manages virtual memory.

3. File Management

Operating system creates, deletes, reads, and writes files and directories.

4. Device Management

OS controls input and output devices like keyboard, printer, mouse, etc.

5. Security and Protection

OS protects system resources and user data from unauthorized access.

6. User Interface

Operating system provides Graphical User Interface (GUI) and Command Line Interface (CLI).

Conclusion

Operating system is the backbone of a computer system because it manages all resources efficiently.

2) Explain OS Services

Operating system provides different services to users and programs for efficient working of computer system.

OS Services

1. Program Execution

OS loads programs into memory and executes them.

2. I/O Operations

Operating system controls input and output devices like keyboard, monitor, and printer.

3. File System Manipulation

OS performs operations such as creating, reading, writing, renaming, and deleting files.

4. Communication

OS allows communication between different processes using shared memory or message passing.

5. Error Detection

Operating system detects hardware and software errors and takes corrective action.

6. Resource Allocation

OS allocates CPU, memory, and devices among multiple users and processes.

7. Security and Protection

Protects system resources and prevents unauthorized access.

Conclusion

OS services make the computer system convenient, secure, and efficient for users.

3) Define System Calls and Explain Types of System Calls

Definition

System calls are requests made by user programs to the operating system kernel to perform specific services.

System calls act as a bridge between user applications and operating system.

Types of System Calls

1. Process Control System Calls

These are used for process creation, execution, and termination.

Examples:

- fork()
- exit()
- wait()

2. File Management System Calls

These are used for file operations such as open, read, write, and close.

Examples:

- open()
- read()
- write()
- close()

3. Device Management System Calls

These are used to access and control hardware devices.

4. Information Maintenance System Calls

These calls obtain or modify system information.

5. Communication System Calls

Used for communication between processes.

Conclusion

System calls provide an interface through which user programs can access operating system services.

UNIT 2

4) Differentiate Between Program and Process

Program	Process
Program is a passive entity	Process is an active entity
Stored on secondary memory	Stored in main memory
Static in nature	Dynamic in nature
Does not have execution state	Has execution state

Conclusion

A process is a program in execution.

5) Explain Process States

A process passes through different states during execution.

Process States

1. New State

Process is being created.

2. Ready State

Process is waiting for CPU allocation.

3. Running State

Process instructions are currently being executed.

4. Waiting/Blocked State

Process waits for I/O operation or event completion.

5. Terminated State

Process execution is completed.

Diagram

New → Ready → Running → Terminated ↓ Waiting ↓ Ready

Conclusion

Process states help operating system manage execution of multiple processes efficiently.

6) Explain Process Control Block (PCB)

Definition

PCB is a data structure maintained by operating system to store information about a process.

Contents of PCB

1. Process ID

Unique identification number of process.

2. Process State

Stores current state of process.

3. Program Counter

Stores address of next instruction to execute.

4. CPU Registers

Stores values of CPU registers.

5. Memory Information

Contains memory allocation details.

6. I/O Information

Stores information about allocated devices and files.

Importance of PCB

- Helps in process scheduling
- Used during context switching
- Maintains process information

Conclusion

PCB is very important for process management in operating system.

7) Difference Between Process and Thread

Process	Thread
Heavyweight	Lightweight
Has separate memory	Shares memory
Communication is slower	Communication is faster
More overhead	Less overhead

Conclusion

Threads improve system performance and responsiveness.

UNIT 3 – Scheduling

8) Explain FCFS Scheduling Algorithm

Definition

FCFS (First Come First Serve) scheduling algorithm executes processes in the order of their arrival.

Advantages

- Simple and easy to implement
- Fair scheduling method

Disadvantages

- Average waiting time is high
- Convoy effect may occur

Conclusion

FCFS is simple but not efficient for time-sharing systems.

9) Explain SJF and SRTF Scheduling

SJF (Shortest Job First)

In SJF scheduling, the process with smallest burst time is executed first.

Advantages

- Minimum average waiting time

Disadvantages

- Starvation may occur for long processes

SRTF (Shortest Remaining Time First)

SRTF is the preemptive version of SJF. The process with shortest remaining burst time gets CPU.

Conclusion

SJF and SRTF improve CPU utilization but may cause starvation.

10) Explain Round Robin Scheduling

Definition

Round Robin scheduling gives each process a fixed CPU time called time quantum.

After time quantum expires, the process moves to ready queue.

Advantages

- Fair scheduling
- Good response time
- Suitable for time-sharing systems

Disadvantages

- More context switching overhead

Conclusion

Round Robin is widely used in modern operating systems.

UNIT 4 – Deadlock

11) Define Deadlock and Explain Coffman Conditions

Definition

Deadlock is a situation in which two or more processes wait forever for resources held by each other.

Coffman Conditions

1. Mutual Exclusion

Only one process can use a resource at a time.

2. Hold and Wait

A process holds one resource and waits for another resource.

3. No Preemption

Resources cannot be taken forcefully from a process.

4. Circular Wait

Processes form a circular chain while waiting for resources.

Conclusion

All four conditions must occur simultaneously for deadlock to happen.

12) Explain Deadlock Prevention Techniques

Deadlock prevention ensures that at least one Coffman condition does not occur.

Techniques

1. Prevent Mutual Exclusion

Use sharable resources whenever possible.

2. Prevent Hold and Wait

Allocate all resources together before execution.

3. Prevent No Preemption

Allow preemption of resources.

4. Prevent Circular Wait

Assign ordering to resources.

Conclusion

Deadlock prevention avoids deadlock but may reduce resource utilization.

13) Explain Banker's Algorithm

Definition

Banker's Algorithm is a deadlock avoidance algorithm used to check whether the system is in safe state or not.

Important Terms

Allocation Matrix

Shows currently allocated resources.

Max Matrix

Shows maximum resources needed.

Need Matrix

$\text{Need} = \text{Max} - \text{Allocation}$

Available

Shows currently available resources.

Safe State

If safe sequence exists, system is in safe state.

Unsafe State

If safe sequence does not exist, system is in unsafe state.

Conclusion

Banker's Algorithm prevents deadlock by maintaining safe state.

UNIT 5 – Memory Management

14) Explain Paging

Definition

Paging is a memory management technique in which logical memory is divided into pages and physical memory is divided into frames.

Pages are loaded into available frames.

Advantages

- Eliminates external fragmentation
- Efficient memory utilization

Disadvantages

- Internal fragmentation may occur

Conclusion

Paging improves memory management efficiency.

15) Difference Between Paging and Segmentation

Paging	Segmentation
Fixed size memory blocks	Variable size memory blocks
Faster technique	Flexible technique
No external fragmentation	External fragmentation possible
Based on physical memory	Based on logical memory

Conclusion

Paging is faster while segmentation provides better logical view.

16) Explain Internal and External Fragmentation

Internal Fragmentation

Unused memory space inside allocated partition is called internal fragmentation.

Example: Partition size = 10 KB Process uses = 8 KB Unused = 2 KB

External Fragmentation

Free memory is available but scattered in small blocks.

Conclusion

Fragmentation reduces memory utilization efficiency.

17) Explain Virtual Memory

Definition

Virtual memory is a technique that allows execution of programs larger than physical memory.

Advantages

- Better memory utilization
- Supports multiprogramming
- Large programs can run easily

Thrashing

Thrashing occurs when excessive page swapping reduces CPU performance.

Conclusion

Virtual memory increases efficiency of memory management.

UNIT 6 – Disk Scheduling

18) Explain FCFS Disk Scheduling

Definition

In FCFS disk scheduling, requests are served in the order of arrival.

Advantages

- Simple and easy
- Fair method

Disadvantages

- High seek time
- Slow performance

Conclusion

FCFS is simple but not efficient.

19) Explain SSTF Disk Scheduling

Definition

SSTF (Shortest Seek Time First) serves the request nearest to current head position.

Advantages

- Less seek time
- Better performance than FCFS

Disadvantages

- Starvation may occur

Conclusion

SSTF improves disk performance.

20) Explain SCAN Disk Scheduling

Definition

In SCAN scheduling, disk arm moves in one direction and serves all requests, then reverses direction.

Why Called Elevator Algorithm?

Because its movement is similar to elevator movement.

Advantages

- Better throughput
- Uniform waiting time

Disadvantages

- Long waiting for some requests

Conclusion

SCAN provides balanced disk scheduling performance.

QUESTIONS HAVING SAME OR SIMILAR ANSWERS

1) OS Functions ↔ OS Services

Questions

- Explain functions of OS
- Explain OS services

Same Answer Parts

- Process management
- Memory management
- File management
- Device management
- Security

Small Change

Functions = Responsibilities of OS Services = Facilities provided by OS

2) Process Concept ↔ Program vs Process

Questions

- Define process
- Explain process concept
- Differentiate between program and process

Same Answer Parts

- Process is program in execution
 - Dynamic nature
 - CPU and memory usage
-

3) PCB ↔ Process Management

Questions

- Explain PCB
- Explain process management

Same Answer Parts

- Process scheduling
 - Process state
 - CPU registers
 - Process information
-

4) SJF ↔ SRTF

Questions

- Explain SJF
- Explain SRTF
- Differentiate between SJF and SRTF

Same Answer Parts

- Shortest burst time
- Waiting time reduction
- Scheduling concept

Small Change

SJF = Non-preemptive SRTF = Preemptive

5) FCFS CPU Scheduling ↔ FCFS Disk Scheduling

Questions

- Explain FCFS scheduling
- Explain FCFS disk scheduling

Same Answer Parts

- First Come First Serve concept
- Arrival order execution

Small Change

CPU scheduling = Processes Disk scheduling = Disk requests

6) Paging ↔ Virtual Memory

Questions

- Explain paging
- Explain virtual memory
- Explain address translation

Same Answer Parts

- Pages and frames
 - Memory management
 - Secondary memory usage
-

7) FIFO ↔ LRU

Questions

- Explain FIFO page replacement
- Explain LRU page replacement
- Find page faults

Same Answer Parts

- Page replacement concept
- Frames
- Page faults

Small Change

FIFO = Oldest page removed LRU = Least recently used page removed

8) Deadlock Conditions ↔ Deadlock Prevention

Questions

- Explain Coffman conditions
- Explain deadlock prevention

Same Answer Parts

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

Small Change

Conditions = Explain conditions Prevention = Explain how to prevent them

9) Multiprogramming ↔ Time Sharing OS

Questions

- Explain multiprogramming OS
- Explain time sharing OS
- Differentiate between them

Same Answer Parts

- Multiple processes
- CPU sharing
- Scheduling

Small Change

Multiprogramming = CPU utilization Time sharing = Fast response time

10) SSTF ↔ SCAN

Questions

- Explain SSTF disk scheduling
- Explain SCAN disk scheduling
- Compare disk scheduling algorithms

Same Answer Parts

- Disk head movement
 - Seek time reduction
-

IMPORTANT SIMILAR ANSWERS (DON'T GET CONFUSED IN EXAM)

1) OS Functions vs OS Services

Similarity

Both explain what operating system does.

Difference

OS Functions	OS Services
Responsibilities of OS	Facilities provided to user
System management side	User interaction side
Example: Memory management	Example: Program execution

Exam Tip

- If question asks “functions”, focus on management work.
 - If question asks “services”, focus on facilities given to users.
-

2) Program vs Process

Program

A set of instructions stored on disk.

Process

A program currently in execution.

Easy Trick

Program = Passive Process = Active

3) Process vs Thread

Process

Independent execution unit with separate memory.

Thread

Small execution unit inside process.

Easy Trick

Thread is lightweight version of process.

4) SJF vs SRTF

SJF	SRTF
Non-preemptive	Preemptive
Once started, process continues	Process can be interrupted
Shortest burst time	Shortest remaining time

Easy Trick

SRTF = Preemptive SJF

5) FCFS CPU Scheduling vs FCFS Disk Scheduling

Similarity

Both work on First Come First Serve basis.

Difference

CPU Scheduling	Disk Scheduling
Schedules CPU processes	Schedules disk requests
Process execution	Disk head movement

6) Paging vs Segmentation

Paging	Segmentation
Fixed size blocks	Variable size blocks
Physical memory view	Logical memory view
Faster	Flexible

Easy Trick

Paging = Equal size Segmentation = Different size

7) Internal vs External Fragmentation

Internal Fragmentation

Unused space inside allocated block.

External Fragmentation

Free space available but scattered.

Easy Trick

Internal = Inside waste External = Outside scattered waste

8) Deadlock vs Starvation

Deadlock	Starvation
Processes wait forever	One process waits long time
No process progresses	Other processes continue

Deadlock	Starvation
Due to circular waiting	Due to unfair scheduling

9) Multiprogramming vs Time Sharing OS

Multiprogramming	Time Sharing
CPU utilization focused	Fast response focused
Multiple jobs in memory	Multiple users share CPU
Non-interactive	Interactive

10) Preemptive vs Non-Preemptive Scheduling

Preemptive	Non-Preemptive
CPU can be taken away	CPU cannot be taken away
Better response time	Simpler
Example: RR	Example: FCFS

IMPORTANT NUMERICALS

1. FCFS Scheduling
2. SRTF Scheduling
3. Round Robin
4. Banker's Algorithm
5. FIFO Page Replacement
6. LRU Page Replacement

EXAM WRITING TIPS

✓ Write definition first ✓ Draw diagrams wherever possible ✓ Write advantages and disadvantages ✓
Underline important keywords ✓ Show formulas clearly in numericals ✓ Write conclusion at end

END